

FreqGAN: Infrared and Visible Image Fusion via Unified Frequency Adversarial Learning

Zhishe Wang^{ID}, *Member, IEEE*, Zhuoqun Zhang, Wuqiang Qi^{ID}, Fengbao Yang, and Jiawei Xu^{ID}

Abstract—Traditional fusion methods based on deep learning mainly employ convolutional or self-attention operations to model local or global dependencies, which often lead to the oversight of frequency-domain information. To address this deficiency, we introduce a unified frequency adversarial learning network, termed FreqGAN. Our method involves a frequency-compensated generator that employs discrete wavelet transformation to decompose encoded spatial features into multiple frequency bands. Leveraging skip connections, low and high-frequency components are respectively directed into the encoder and decoder, compensating for additional outline and detail. Moreover, we construct a hybrid frequency aggregation module, which enables a progressive optimization of activity levels across multiple scales and makes the various frequency bands correlated. Complementing our generative model, we devise dual frequency-constrained discriminators. These discriminators are tasked with dynamically adjusting weights for each input frequency band, thereby obligating the generator to accurately reconstruct salient frequency information from different modality images. Additionally, a frequency-supervised function is formulated to further safeguard against the loss of frequency information. Our comprehensive experimental evaluations, encompassing a wide range of fusion tasks and subsequent applications, distinctly highlight FreqGAN's superior performance, establishing it as a frontrunner in comparison to existing state-of-the-art alternatives. The source codes are forthcoming at: <https://github.com/Zhishe-Wang/FreqGAN>.

Index Terms—Image fusion, wavelet transformation, hybrid frequency aggregation, generative adversarial network.

I. INTRODUCTION

IMAGE fusion plays an indispensable role in integrating disparate yet synergistic information from multiple sensors into a unified representation. Typically, visible sensors excel in

capturing high-resolution scene details through light reflection imaging. However, their performance is notably compromised under suboptimal conditions such as poor weather or obscured environments. Conversely, infrared sensors are skilled in detecting thermal emissions from objects, thereby delineating object features amidst complex backgrounds. Image fusion technology harnesses the distinct imaging capabilities of these two sensor types, synthesizing their complementary strengths to produce images that are both detailed and comprehensive. This process effectively mitigates the limitations that are inherent when using each sensor independently. Furthermore, the application of fused results has proven to significantly enhance the efficacy of some downstream tasks, such as pedestrian recognition [1] and RGB-T tracking [2].

Over recent decades, a variety of traditional methodologies have been developed to address the challenges inherent in infrared and visible image fusion. These paradigms include approaches based on multi-scale transform [3], sparse representation [4], optimization estimation [5], hybrid models [6]. Typically, these methods rely on mathematical models for feature extraction or optimization. However, a significant limitation of these approaches is their reliance on a uniform framework for feature extraction. This generalization overlooks the distinct characteristics of features unique to different modalities, thereby limiting their applicability across varied scenarios. Furthermore, the utilization of hand-crafted fusion rules or subjective activity level measurement introduces an element of bias, potentially compromising the objectivity and reliability of the fusion outputs. Such biases can significantly detract from the utility of the fused images for objective analysis and decision-making purposes.

In recent years, the domain of deep learning has witnessed substantial advancements, primarily attributed to its robust capabilities in feature extraction and representation. This field encompasses a variety of prevalent methodologies, including autoencoder (AE)-based [7], [8], convolutional neural network (CNN)-based [9], [10], Transformer-based [11], [12], generative adversarial network (GAN)-based [13], [14], and other models [15]. AE-based frameworks typically employ a shared encoder-decoder architecture for feature extraction and feature reconstruction, integrating these features through a specially designed fusion layer. CNN-based approaches frequently utilize either image-level or feature-level fusion schemes, which involve the concatenation of source images or the aggregation of deep features at various stages. Transformer-based models, distinct from the aforementioned approaches,

Received 22 February 2024; revised 3 August 2024; accepted 10 September 2024. Date of publication 13 September 2024; date of current version 30 January 2025. This work was supported in part by the Fundamental Research Program of Shanxi Province under Grant 202203021221144, in part by the Patent Transformation Program of Shanxi Province under Grant 202405012, and in part by Zhejiang Provincial Natural Science Foundation under Grant LY23F030001. This article was recommended by Associate Editor A. Iosifidis. (*Corresponding author: Jiawei Xu.*)

Zhishe Wang, Zhuoqun Zhang, and Wuqiang Qi are with the School of Applied Science, Taiyuan University of Science and Technology, Taiyuan 030024, China (e-mail: wangzs@tyust.edu.cn; zhuoqunzhang001@163.com; qwqqwq@stu.tyust.edu.cn).

Fengbao Yang is with the School of Information and Communication Engineering, North University of China, Taiyuan 030051, China (e-mail: yangfb@nuc.edu.cn).

Jiawei Xu is with the Key Laboratory of Intelligent Informatics for Safety and Emergency of Zhejiang Province, Wenzhou University, Wenzhou 325035, China (e-mail: jxulincoln@gmail.com).

Digital Object Identifier 10.1109/TCSVT.2024.3460172

1051-8215 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See <https://www.ieee.org/publications/rights/index.html> for more information.

capitalize on self-attention mechanisms to effectively capture long-range dependencies, thus comprehending global contexts. In contrast, GAN-based methods utilize adversarial training strategies to ensure that the distribution of fused outputs aligns closely with that of source images.

Although current fusion approaches based on deep learning produce impressive results, a notable oversight is that frequency-domain information is not taken into account. As elucidated by the F-principal [16], neural networks exhibit a tendency to first focus quickly on capturing low-frequency (LF) signals and then shift their attention slowly to high-frequency (HF) details while keeping them small. This indicates that the forward propagation of fusion network often causes the loss of frequency information. To this end, our work aims to explore a more reasonable paradigm from a unified frequency-domain perspective. This frequency-centric perspective is critical for a comprehensive understanding of feature representation in deep learning-based image fusion.

Fig.1 visualizes the LF and HF components of an infrared and visible image pair via discrete wavelet transformation (DWT). The LF signals encompass general information such as overall background and structural outlines, signifying the modality-shared information. Conversely, the HF components encapsulate rich details and perceptible information like house textures and pedestrian details, representing their modality-specific information. As shown in Fig.1 (d), (h), and (i), compared with its variant without frequency information, designated as w/o Freq, FreqGAN achieves not only optimal brightness control but also maintains superior detail clarity. Meanwhile, its quantitative performance comprehensively surpasses that of w/o Freq on eight metrics. Therefore, the considerations of frequency components can cater to the intrinsic learning mechanism of the fusion network, and facilitate it to effectively compensate for the common and unique characteristics of different modalities without specialized design.

Based on the above analysis, this paper introduces FreqGAN, a meticulously engineered unified frequency adversarial learning network. Initially, we employ DWT to segregate encoded spatial features into multiple frequency bands. Utilizing skip connections, LF signals and HF components are fed into the encoder and decoder, respectively. Meanwhile, we develop a hybrid frequency aggregation module to progressively optimize the activity levels of various frequency bands, effectively capturing the unique characteristics of cross-modality features. These strategies can explicitly and directly leverage multiple frequency bands to provide compensatory structural content and edge details. Furthermore, we introduce dual frequency-constrained discriminators that leverage DWT to transform both source images and fused outputs into diverse frequency components. Their dynamic learning mechanism assigns tailored weights to each input component, ensuring the generator accurately reconstructs essential frequency information from different modality images. Finally, we incorporate a frequency-supervised function to further prevent the loss of frequency content in terms of structural similarity, detail preservation, and intensity control. To summarize, FreqGAN encapsulates four significant contributions.

- We devise a frequency-compensated generator that executes DWT to decompose encoded spatial features into multiple frequency bands. The low and high-frequency components are fed into the corresponding encoder and decoder to provide extra outlines and details.
- We construct a hybrid frequency aggregation module to integrate signals across fine to coarse scales. This process leverages prior fusion outputs to calculate attentional weights for frequency components, and progressively optimize the activity levels at varying scales.
- We develop dual frequency-constrained discriminators to facilitate adaptive learning of frequency components, which can dynamically learn weights for each input component and force the generator to reconstruct crucial frequency information from cross-modality images.
- We formulate a frequency-supervised function to further prevent the loss of frequency information. This meticulous design drives the fusion network towards preserving abundant texture details and structural information while achieving optimal intensity control.

The remainder of this paper is organized as follows. Section II mainly describes the development of fusion technology. Section III elaborates on the overall framework of our FreqGAN. Section IV and Section V present experimental comparisons and draw relevant conclusions, respectively.

II. RELATED WORK

This section offers a comprehensive overview of infrared and visible image fusion. Meanwhile, frequency-domain learning strategies are also presented.

A. CNN-Based Methods

Convolutional neural networks have significantly advanced image fusion tasks, primarily attributed to their robust capacity for feature representation. Li and Wu introduced DenseFuse [7], which innovatively integrated skip connections into encoder-decoder networks, enhancing feature representation capabilities. In a bid to further augment performance, Wang et al. developed the Res2Fusion [8] and UNFusion [17] methods, utilizing attention-based mechanisms to refine and amalgamate features. Despite their advancements, these methodologies are still reliant on manually crafted fusion strategies, limiting their practical applicability.

In response to the constraints posed by manual design approaches, Li et al. proposed LRRNet [9], which established a mathematical correlation between optimal solution and network framework, thereby obviating the need for empirical design. Yang et al. introduced a multi-scale dual attention framework, termed MDA [10], which integrated complementary information from the image and patch level. Expanding the application spectrum to high-level tasks, Tang et al. presented PIAFusion [18], which took into account image illumination conditions to enhance object detection and semantic segmentation. Wang et al. developed a novel approach for cross-modality image generation and registration [19]. However, a common limitation among these methods is their adherence to the fundamental convolutional principle for

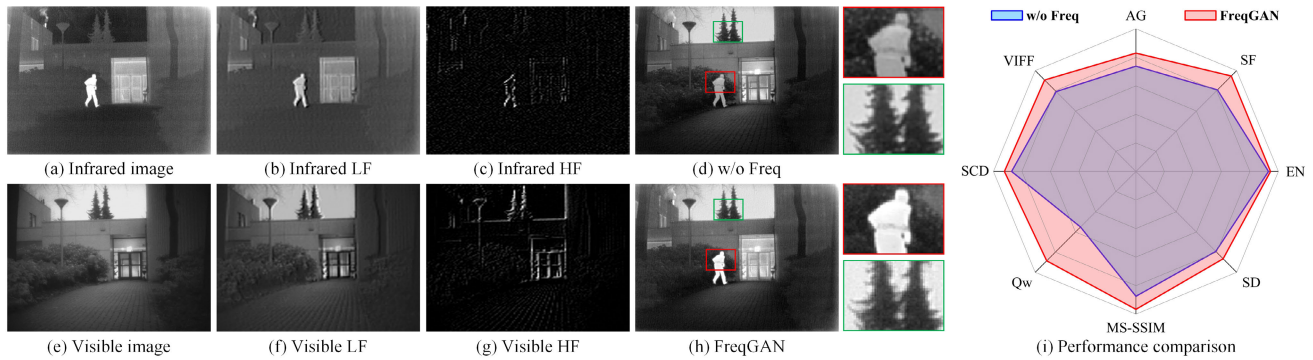


Fig. 1. Visualization of frequency components. LF and HF respectively contain modality-shared and modality-specific information. Note that FreqGAN maintains optimal brightness control and superior detail clarity compared with its variant without frequency information.

establishing local features. This process potentially neglects the modeling of long-range dependencies, thereby risking the omission of vital contextual content.

B. Transformer-Based Methods

As an alternative to convolution operations, the self-attention mechanism emerges as a potent tool for long-range modeling, greatly enhancing feature representation and thus bolstering fusion performance. Illustrating this, Wang et al. introduced SwinFuse [11], which innovated with a fully attentional feature encoding network. Similarly, Zhao et al. proposed a dual-branch Transformer-CNN encoder, adept at extracting both global and local features, and devised tailored fusion strategies [20]. These methodologies are reminiscent of the above AE-based models, and still necessitate manually crafted fusion rules.

To achieve end-to-end self-supervised learning, Tang et al. introduced DATFuse [12], proposing distinct serial structures for the aggregation of local and global features. Ma et al. unveiled SwinFusion [21]. This approach was distinguished by its focus on both inter-domain and cross-domain global learning, applicable to multiple fusion tasks. Li and Wu presented a novel cross attention fusion framework, namely CrossFuse [22], which employed self-attention and cross-attention to strengthen the intra-feature and inter-feature, respectively. Wang et al. developed a collaborative learning scheme of both CNN and Transformer to alternately incorporate local and global relationships [23]. Moreover, Liu et al. presented SegMiF [24], which established a semantic/modality multi-head attention mechanism, applicable to both image fusion and segmentation. These methods collectively excel in synthesizing local and global relationships to enhance fusion performance. However, a notable challenge inherent in the fusion domain is the lack of ground truth, which restricts these models from fully realizing their potential. This limitation underscores the need for developing methodologies that can navigate the unique complexities of fusion tasks without relying on conventional ground truth benchmarks.

C. GAN-Based Methods

Distinct from the aforementioned learning paradigms, GAN-based methodologies conceptualize image fusion as a

generative task. Ma et al. first presented FusionGAN [13], which underscored the transformation of varied image distributions into a unified output. Since a singular discriminator inadvertently led to a bias towards infrared imagery, they introduced a dual-discriminator design in subsequent work [14], meticulously engineered to enhance the retention of visible details in the fused output. These methods collectively focus on discriminator optimization as a means to guide the generator towards producing high-quality fused outputs.

In parallel, there has been a significant focus on generator architectures. Yang et al. integrated a squeeze-and-excitation model into their generator design to accentuate texture information [25]. Wang et al. introduced ICAFusion [26], which employed a tripartite path in the generator alongside dual attentional modules, facilitating the efficient transfer and compensation of attention features. Additionally, they innovated a cross-scale iterative interaction within the generator to progressively optimize activity levels [27]. Addressing diverse application requirements, Liu et al. proposed a target-aware GAN framework for both image fusion and object detection [28]. However, these methods aim to design flexible discriminator or generator architectures. The core purpose is to facilitate feature representation ability and achieve a better fusion effect. However, these methods overlook specialized frequency information, potentially affecting fusion effectiveness.

D. Frequency-Domain Learning

Recently, some researchers have introduced the frequency transformation characteristics into neural network models for image fusion tasks. More specifically, Yao et al. employed the Laplacian Pyramid to decompose source images, and then proposed different sub-networks for the high- and low-frequency components [29]. Li et al. introduced a decomposition pooling-based fusion network, which exploited a de-pooling layer instead of the conventional max-pooling techniques to prevent information loss [30]. Zheng et al. probed a synergistic high-order interaction network and incorporated fast Fourier transform (FFT) into the self-attention mechanism to simplify matrix multiplication [31]. These methods improve the fusion performance from the single perspective of frequency decomposition or enhancement.

Different from the above efforts, we design FreqGAN from a unified frequency-domain perspective to confront the

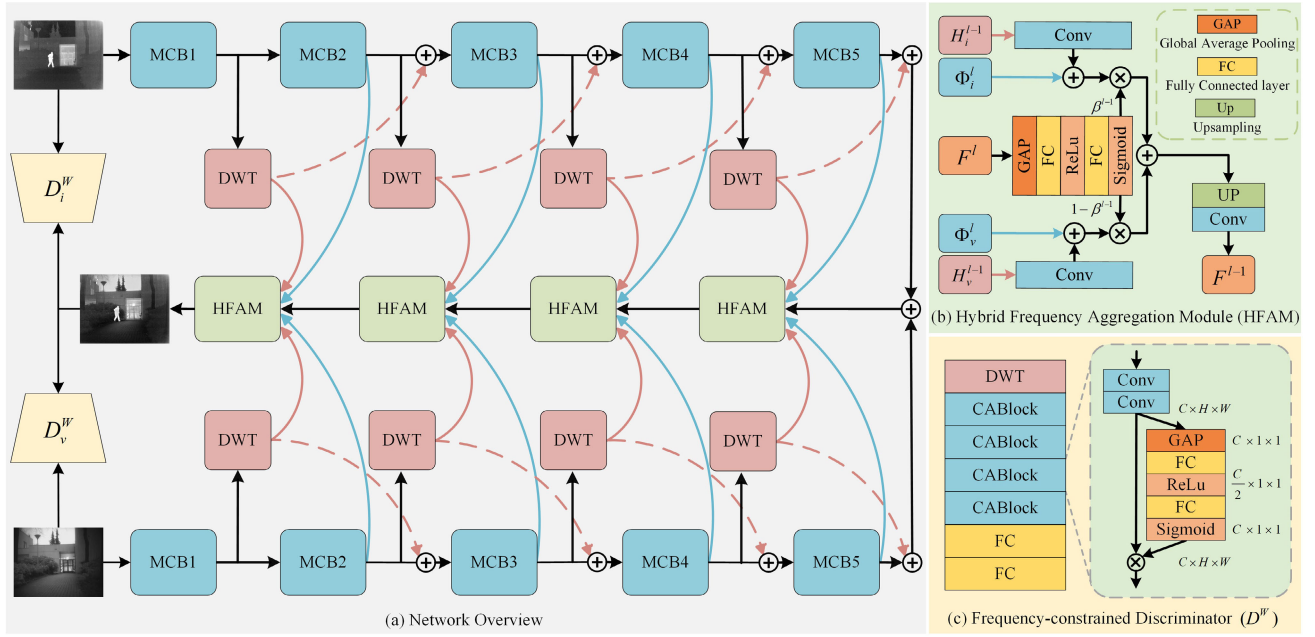


Fig. 2. The overall framework of our FreqGAN. Source images are fed into multi-level convolutional blocks (MCBs) to generate encoded spatial features. These features are further decomposed by DWT into different frequency components. Employing skip connections, low-frequency signals are directed back into the encoder, whereas high-frequency components are integrated with subsequent encoded features using hybrid frequency aggregation modules (HFAMs).

influence of frequency information on image fusion, involving the generator, discriminator, and loss function of the GAN framework. Specifically, our novel network anchors in the strategic manipulation of frequency components, the refinement of feature aggregation techniques, and the enforcement of stringent fidelity constraints, aiming to address this overlooked aspect for improved fusion quality.

III. METHODOLOGY

This section describes the overall framework that consists of a frequency-compensated generator, frequency-constrained discriminators, and a frequency-supervised loss function.

A. Frequency-Compensated Generator

As illustrated in Fig.2(a), the generator introduces a dual-branch encoder and a decoder. Each encoder branch consists of five multi-level convolutional blocks, labeled MCB1, MCB2, MCB3, MCB4, and MCB5, alongside four discrete wavelet transformation blocks. Each MCB comprises two convolution layers utilizing 3×3 kernel size, with strides set to 1 and 2, respectively. The filter banks at different scales are set to 16, 32, 64, 128, and 256. All activation layers are implemented by PReLU functions. To disentangle the encoder features into multiple frequency components, we adopt the Haar wavelet with four kernels, namely LL^T , LH^T , HL^T , and HH^T , which are defined by Eq.1.

$$L^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad H^T = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \end{bmatrix} \quad (1)$$

where L and H represent the low-pass and high-pass filters. In particular, for the encoded infrared and visible features Φ_i^l and Φ_v^l extracted by l -th convolutional block, where l denotes the scaling factor, Haar wavelet transforms them into

frequency components. Employing skip connections, we compensate the low-frequency signals for the subsequent encoded features Φ_i^{l+1} and Φ_v^{l+1} obtained by the $(l+1)$ -th convolution blocks, which are formulated by Eq.2 and Eq.3, respectively.

$$\Phi_i^{l+2} = LL_i^l + MCB(\Phi_i^{l+1}) \quad (2)$$

$$\Phi_v^{l+2} = LL_v^l + MCB(\Phi_v^{l+1}) \quad (3)$$

The features Φ_i^{l+2} and Φ_v^{l+2} are then fed into the $(l+2)$ -th convolution block. Thus, the low-frequency skip connections contribute extra structural outlines to the encoded features.

Moreover, the decoder comprises four hybrid frequency aggregation modules (HFAMs) that integrate high-frequency components and encoded features to produce the fused outputs. As shown in Fig.2(b), leveraging prior fused output F^l , we employ a squeeze-and-excitation block [32] to calculate attentional weights β^{l-1} , which is described by Eq.4.

$$\beta^{l-1} = SE(F^{l-1}) \quad (4)$$

where $SE(\cdot)$ denotes the SE block operation that produces channel-wise attentional weights. Subsequently, we perform the tensor addition on all the high-frequency components, i.e., $H = LH + HL + HH$, and employ a 3×3 convolution layer to adjust their channel numbers. Meanwhile, we assign the weights β^{l-1} for the activity levels of the infrared path and $1 - \beta^{l-1}$ for the visible path. After an upsampling operation and a 3×3 convolution layer, the generated fused outputs can be obtained, which are simply formulated by Eq.5.

$$F^{l-1} = \beta^{l-1} \cdot (\Phi_i^l + H_i^{l-1}) + (1 - \beta^{l-1}) \cdot (\Phi_v^l + H_v^{l-1}) \quad (5)$$

where H_i^{l-1} and H_v^{l-1} denotes the summation of all the high-frequency components for infrared and visible images. Subsequently, the generated fused outputs are fed into the next

HFAM, enabling the gradual optimization of activity levels to produce high-quality fusion images.

B. Frequency-Constrained Discriminator

In the realm of GAN-based fusion models, the discriminator traditionally functions to discern spatial data distributions between fused outputs and source images. However, source images often contain elements of invalid information, such as irrelevant infrared background content. A reliance on spatial distribution as the sole discriminative criterion may inadvertently result in the undue retention of such non-essential information. Additionally, the complex dynamics governing the interaction between the generator and discriminator can amplify the influence of these undesired elements, potentially compromising the fidelity of critical modality characteristics. To address these issues, we conceptualize and implement dual frequency-constrained discriminators, designated as D_i^W and D_v^W . Uniquely, these discriminators operate on the principle of the frequency domain, incorporating channel attention blocks designed to selectively suppress unwanted information.

As depicted in Fig.2(c), our approach commences with the application of discrete wavelet transformation to both the fused outputs and source images. This process decomposes the image data into low-frequency signals and high-frequency components. Subsequently, these frequency bands are concatenated and subjected to a series of convolutional attention blocks (CABlocks) to compute category scores. Each CABlock is composed of dual convolution layers, employing a 3×3 kernel size, with strides meticulously set at 1 and 2. The input and output channel configurations for these blocks are strategically arranged as 4:8, 8:16, 16:32, and 32:64. The primary function of these CABlocks is to extract and refine features from the four distinct frequency components. Acknowledging the varying significance of each frequency component in defining object outlines and background details, we integrate a squeeze-and-excitation model within our discriminators. This model is adept at assigning weights to the features extracted from various frequency bands, thus enhancing the overall quality.

C. Frequency-Supervised Loss Function

In the proposed FreqGAN, we formulate the loss function for both the generator and dual discriminators in the frequency domain. Thus, within the generator, the loss function (L_G) is composed of the wavelet content loss (L_{con}^W) and the wavelet adversarial loss (L_{adv}^W), which is defined by Eq.6.

$$L_G = L_{con}^W + L_{adv}^W \quad (6)$$

Inspired by SwinFusion [21], we extend the structural similarity loss, texture loss, and intensity loss by combining them with discrete wavelet transformation. Concretely, the wavelet SSIM loss (L_{ssim}^W) is formulated by Eq.7.

$$L_{ssim}^W = \sum_0^w \frac{r_w}{2} ((1 - ssim(I_i^w, I_f^w)) + (1 - ssim(I_v^w, I_f^w))) \quad (7)$$

where $ssim(\cdot)$ represents the structural similarity operation. I_f^w , I_i^w , and I_v^w denote different frequency components of fused output and source images. Additionally, $w = \{0, 1, 2, 3\}$ represents the index of frequency components $\{LL, HL, LH, HH\}$. The parameter r_w , which controls the weights of each component, is defined by Eq.8.

$$LL : LH : HL : HH = r^2 : r(1 - r) : r(1 - r) : (1 - r)^2 \quad (8)$$

Furthermore, the wavelet texture loss (L_{grad}^W) and wavelet intensity loss (L_{int}^W) compel the network to diligently retain texture details and intensity information from both modalities, which are formulated by Eq.9 and Eq.10.

$$L_{grad}^W = \sum_0^w \frac{r_w}{HW} \left\| |\nabla I_f^w| - \max(|\nabla I_i^w|, |\nabla I_v^w|) \right\|_1 \quad (9)$$

$$L_{int}^W = \sum_0^w \frac{r_w}{HW} \left\| I_f^w - \text{mean}(I_i^w, I_v^w) \right\|_1 \quad (10)$$

where H and W denote the height and width of source images, $\max(\cdot)$ and $\text{mean}(\cdot)$ signify the maximum and average selection, respectively. Meanwhile, $|\cdot|_1$ and ∇ stand for L1-norm operation and Sobel gradient operator, respectively.

Thus, the wavelet content loss function can be formulated by Eq.11.

$$L_{con}^W = \lambda_1 L_{ssim}^W + \lambda_2 L_{grad}^W + \lambda_3 L_{int}^W \quad (11)$$

where λ_1 , λ_2 , and λ_3 control the balance between three frequency loss functions.

In the dual discriminators, infrared discriminator (D_i^W) and visible discriminator (D_v^W) are proposed to distinguish data distribution across different frequency components. To be specific, the adversarial loss function is formulated by Eq.12.

$$L_{adv}^W = -\frac{1}{N} \sum_{n=1}^N ([D_i^W(I_f^{w,n})] + [D_v^W(I_f^{w,n})]) \quad (12)$$

Additionally, the frequency loss functions of dual discriminators are formulated by Eq.13 and Eq.14, respectively.

$$L_{D_i^W} = \frac{1}{N} \sum_{n=1}^N [D_i^W(I_{i,f}^{w,n}) + \gamma (\|\nabla D_i^W(I_{i,f}^{w,n})\|_2 - 1)^2] \quad (13)$$

$$L_{D_v^W} = \frac{1}{N} \sum_{n=1}^N [D_v^W(I_{v,f}^{w,n}) + \gamma (\|\nabla D_v^W(I_{v,f}^{w,n})\|_2 - 1)^2] \quad (14)$$

where $\|\cdot\|_2$ and γ signify the L2-norm operation and regularization parameter. The first term accounts for the Wasserstein distance between fused output and infrared or visible input. The second term denotes the gradient penalty, serving to constrain the learning capacity of dual discriminators.

IV. EXPERIMENTS AND DISCUSSIONS

This section first describes the experimental configurations during the training and testing phases. Following this, fusion experiments and downstream applications are conducted to ascertain the effectiveness of FreqGAN. Finally, ablation studies are outlined to provide further insights into our method.

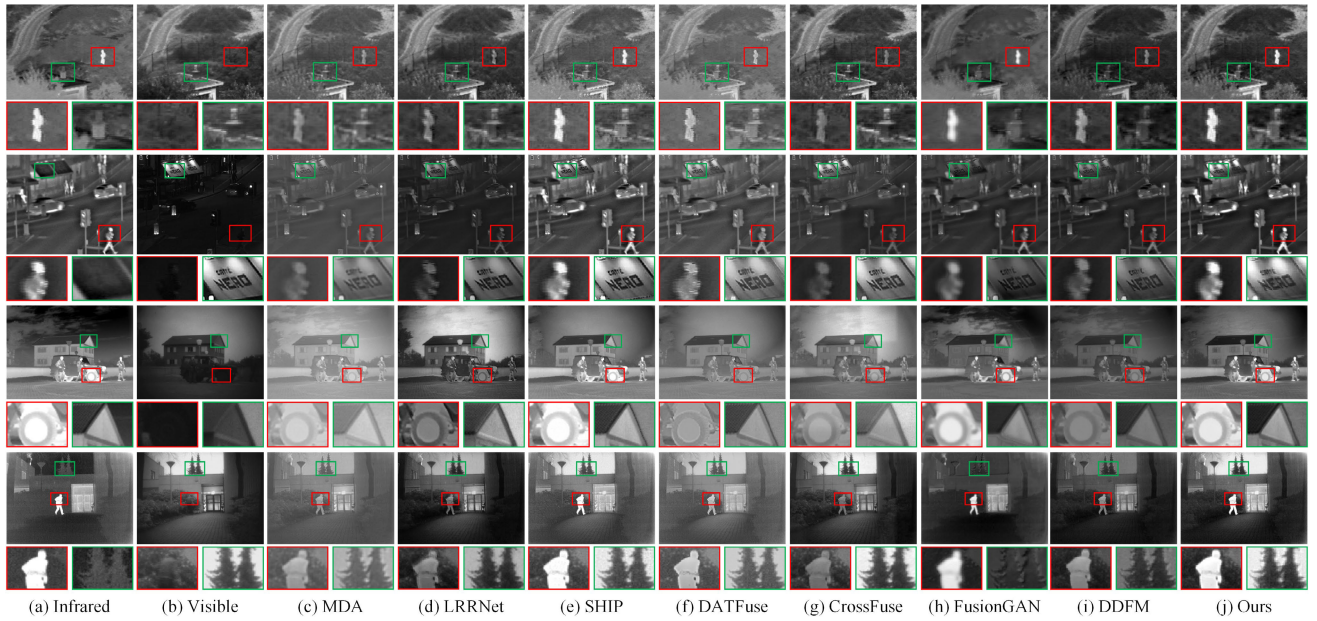


Fig. 3. Qualitative descriptions of our FreqGAN with the other seven SOTA methods on the TNO benchmark. Specially, four representative examples, namely *Nato_camp*, *Street*, *Soldiers_with_jeep*, and *Kaptein_1123*, are selected for a comprehensive visual comparison.

A. Experimental Configurations

During the training phase, we employ the TNO¹ benchmark to crop 256×256 image patches with a sliding step of 12. The grayscale values of these patches are normalized into the range of $[0, 1]$. This process yields a total of 10563 image pairs used for model training. Meanwhile, we adopt the Adam optimizer with a learning rate of 1×10^{-4} for updating the training model. The batch size and epoch are set to 4 and 8, respectively. Empirically, the hyperparameters of the loss function λ_1 , λ_2 , and λ_3 , are set to 10, 16, and 14, respectively. The control parameter of frequency component r is set to 0.4. The experimental platform proposes an Intel I9-10850K, 64 GB memory, and an NVIDIA GeForce GTX 3090 GPU.

During the testing phase, we conduct experimental validations on both TNO and M³FD² benchmarks. The corresponding 25 and 40 image pairs are selected from these two datasets to assess fusion performance. Meanwhile, the downstream application tasks include object detection on M³FD benchmark and semantic segmentation on FMB³ benchmark. Furthermore, in our comparative analysis, we consider seven state-of-the-art methods, such as MDA [10], LRRNet [9], SHIP [31], DATFuse [12], CrossFuse [22], FusionGAN [13], DDFM [15]. Additionally, to evaluate the performance, we employ eight metrics, such as average gradient (AG), spatial frequency (SF) [33], entropy (EN) [34], standard deviation (SD) [35], multi-scale structural similarity (MS-SSIM) [36], Qw [37], the sum of the correlation differences (SCD) [38], and visual information fidelity for fusion (VIFF) [39].

B. Results on TNO Benchmark

To substantiate the efficacy of FreqGAN, we commence our experimental analysis with the TNO benchmark, and select four illustrative examples, namely *Nato_camp*, *Street*, *Soldiers_with_jeep*, and *Kaptein_1123*, to conduct a detailed visual comparison. The qualitative results of these examples are presented in Fig.3. For an enhanced visual examination, we demarcate local infrared targets and visible details using red and green boxes, respectively. From the results, it is discernible that CNN-based methods, such as MDA and LRRNet, predominantly concentrate on local feature modeling, often at the expense of essential object brightness and background detail preservation. Transformer-based approaches like DATFuse and CorssFuse exhibit proficiency in aggregating local and global features. However, they are not immune to issues like diminished brightness and loss of fine details. SHIP, through its synergistic high-order interaction, demonstrates an improved performance yet still faces challenges in concurrently preserving target brightness and visible detail fidelity. FusionGAN, while prioritizing object information retention, tends to compromise on the clarity of visible details. DDFM, with its integration of inference solution and diffusion sampling in a unified iterative framework, shows promise in image fusion but lacks efficacy in effectively capturing thermal radiation. In stark contrast, FreqGAN outperforms these methods, achieving not only optimal brightness control but also maintaining superior scene detail clarity.

For a comprehensive quantitative assessment, we employ eight previously mentioned metrics to evaluate fusion performance. Fig.4 delineates the comparative results between FreqGAN and seven other competing methods. The performance curve of our FreqGAN is indicated by a red dotted line, with the red bold and blue underlined figures denoting the optimal and suboptimal values, respectively. FreqGAN

¹[Online]. Available: https://figshare.com/articles/TN_Image_Fusion_Dataset/1008029

²[Online]. Available: <https://github.com/dlut-dimt/TarDAL>

³[Online]. Available: <https://github.com/JinyuanLiu-CV/SegMiF>

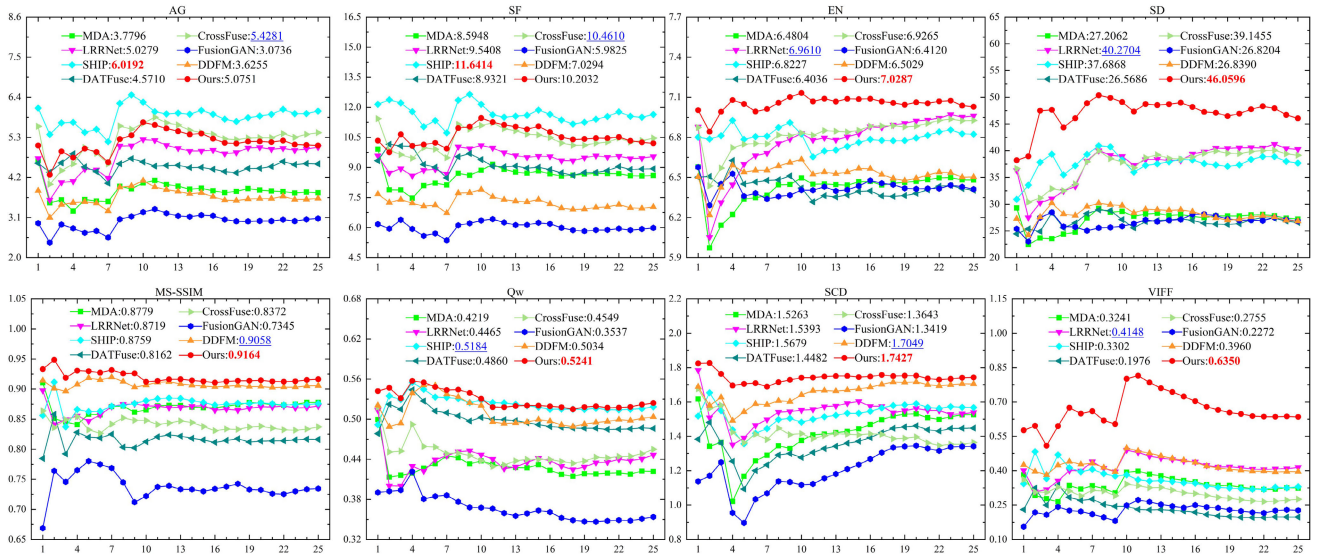


Fig. 4. Quantitative comparison results of our FreqGAN with the other seven SOTA methods on the TNO benchmark. Our FreqGAN (indicated by the red dotted line) secures the top rank for metrics EN, SD, MS-SSIM, Qw, SCD and VIFF, while clinching the third position for AG and SF.



Fig. 5. Qualitative descriptions of our FreqGAN with the other seven SOTA methods on the M^3FD benchmark. Specially, four representative examples, namely 00232, 00025, 03878 and 00762, are selected for a comprehensive visual comparison.

consistently achieves top-ranking positions across multiple metrics, including EN, SD, MS-SSIM, Qw, SCD, and VIFF, while securing third place in AG and SF. These results collectively highlight the remarkable capabilities of FreqGAN, particularly in terms of clarity and visual impact, as evidenced by its significantly higher scores in SD and VIFF. The attainment of optimal values in EN and SCD reinforces FreqGAN's ability to transfer a wealth of information to the fused output while maintaining a strong correlation with the source images. Additionally, the optimal MS-SSIM and Qw scores further attest to FreqGAN's advantage in preserving edge and structural integrity. In summary, these quantitative experiments robustly validate the superiority of FreqGAN.

C. Results on M^3FD Benchmark

We further expand our assessment by conducting experiments on the M^3FD benchmark, rigorously comparing FreqGAN against other competitors to gauge its generalization ability. Specifically focusing on color image fusion tasks, our method involves the initial transformation of the RGB visible image into the YCbCr color space. Following this process, the proposed fusion model seamlessly integrates the Y channel with the infrared image. Then, the generated fusion image is restored to the original RGB format. Fig. 5 presents the qualitative comparison results of four examples, namely 00232, 00025, 03878, and 00762. It can be found that our FreqGAN demonstrates significant advantages in controlling

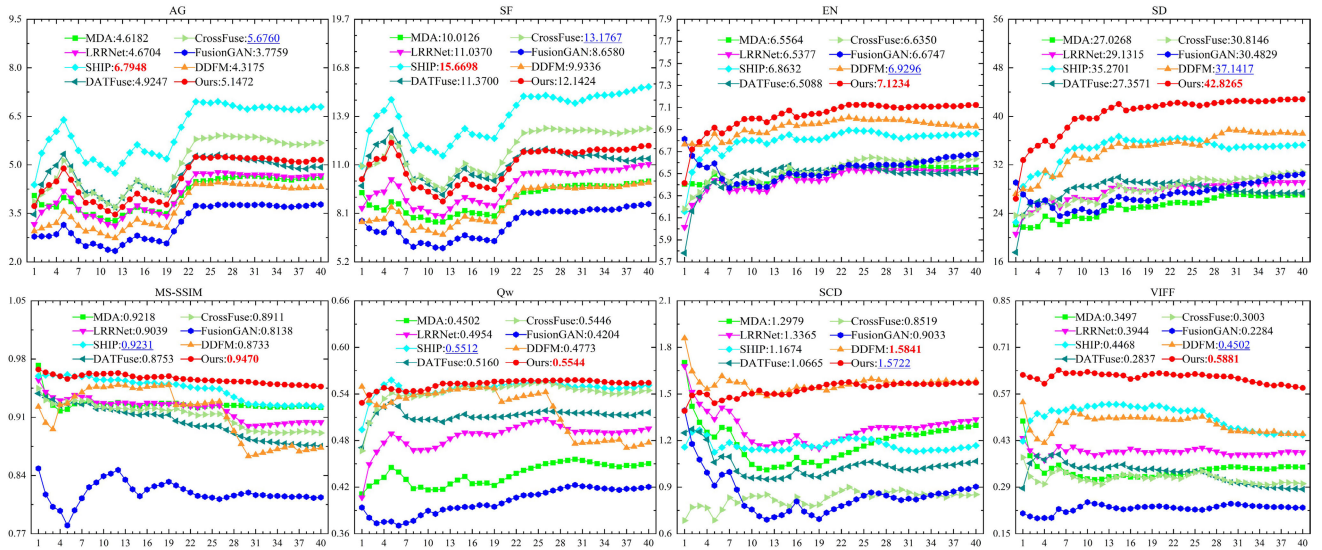


Fig. 6. Quantitative comparison results of our FreqGAN with the other seven SOTA methods on the M³FD benchmark. Our FreqGAN (indicated by the red dotted line) attains the top ranking across EN, SD, MS-SSIM, Qw and VIFF, holds the second position for SCD, and ranks third for AG and SF.

intensity and preserving details. For salient pedestrian objects, FreqGAN retains high brightness characteristics and clear edge contours. Additionally, it accurately describes background details like trees, high-voltage towers, and overhead girders with exceptional clarity. Furthermore, Fig. 6 depicts their quantitative comparison results. The proposed FreqGAN attains the top ranking across EN, SD, MS-SSIM, Qw, and VIFF metrics, holds the second position for SCD, and ranks third for AG and SF. Both qualitative and quantitative experiments unequivocally demonstrate that our FreqGAN outperforms other seven state-of-the-art competitors, showcasing its promising fusion performance and excellent generalization ability.

The experiments are conducted on both TNO and M³FD benchmarks to unequivocally establish the superior performance of our FreqGAN across diverse illumination scenes and object categories. Several key factors contribute to this preponderance. First of all, the generator utilizes skip connections to respectively incorporate low-frequency and high-frequency signals into the encoder and decoder. The frequency components compensate for additional outline and detail information, enhancing the overall effect. Moreover, the hybrid frequency aggregation module is specifically engineered to optimize activity levels and capture the intrinsic characteristics presented in different modality images. Secondly, the development of dual frequency-constrained discriminators plays a pivotal role in our model. These discriminators enable adaptive learning of frequency components, effectively distinguishing differences from the generator and compelling it to accurately reconstruct crucial frequency information. Finally, our frequency-supervised loss function leverages structural similarity, detail preservation, and intensity control to further safeguard against the loss of frequency information. Consequently, our FreqGAN ensures that the fused outputs prioritize infrared salient objects while preserving vital visible details, contributing to its superior performance.

D. Downstream Application

To thoroughly assess the practical efficacy of our proposed method, we extended our evaluation to downstream applications, notably object detection and semantic segmentation, to elucidate the beneficial impact of image fusion.

Image Fusion for Object Detection: Our investigation into the influence of image fusion on object detection commences with an analysis using the M³FD dataset. This benchmark encompasses 4,200 images annotated with 33,603 objects across six categories. We employ the YOLOv5 network [40] as the detection model. The performance metrics center on mean average precision (mAP), where mAP@0.5 delineates an accuracy threshold of 0.5 at an intersection-over-union (IoU) metric, and mAP@[0.5:0.95] signifies the averaged performance over IoU thresholds from 0.5 to 0.95, with an increment of 0.05. For a balanced analysis, the detection model is applied to both source images and fused outputs.

Qualitative comparisons of detection performance are depicted in Fig. 7. Notably, for prominent objects like pedestrians and cars, our FreqGAN demonstrates superior precision values in comparison to both the source images and other competitors. This indicates that our fused outputs more effectively facilitate object detection applications. Table I presents the quantitatively comparative results. Most fusion models exhibit robust detection capabilities, with their average precision values markedly surpassing those obtained using solely infrared or visible images. Remarkably, our FreqGAN outshines other competitors, registering improvements of 3.41% and 3.60% in mAP@0.5 and mAP@[0.5:0.95], respectively. This phenomenon underscores FreqGAN's capacity to meticulously extract and integrate complementary characteristics from different modality images, thereby endowing the detector with enhanced performance.

Image Fusion for Semantic Segmentation: We extend our validation of the proposed FreqGAN to the domain of semantic segmentation. For this purpose, we utilize the full-time

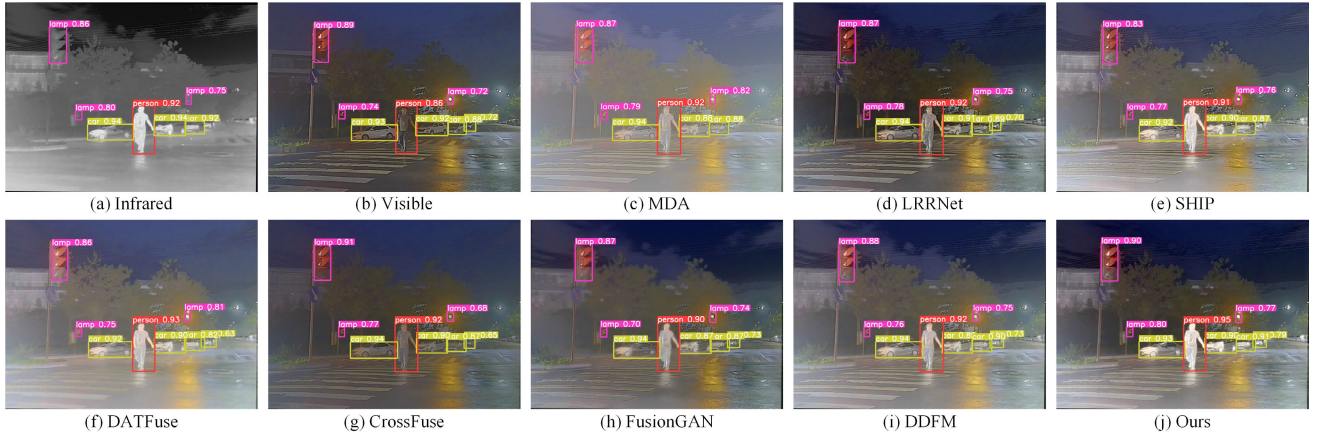


Fig. 7. Qualitative comparisons of detection performance for our FreqGAN with the other seven SOTA methods and source images on the M³FD benchmark. Note that our FreqGAN exhibits higher precision values for different object categories.

TABLE I
QUANTITATIVE COMPARISONS OF DETECTION PERFORMANCE ON THE M³FD BENCHMARK

Methods	mAP@0.5							mAP@[0.5:0.95]						
	Person	Car	Bus	Lamp	Motorcycle	Truck	All	Person	Car	Bus	Lamp	Motorcycle	Truck	All
Infrared	0.807	0.924	<u>0.963</u>	0.621	0.698	0.891	0.817	0.465	0.648	0.764	0.263	0.401	0.612	0.526
Visible	0.691	<u>0.936</u>	0.960	<u>0.830</u>	0.761	<u>0.921</u>	0.850	0.356	0.672	0.767	0.415	0.437	0.646	0.549
MDA [10]	0.782	0.900	0.941	0.813	0.789	0.890	0.852	0.459	0.650	0.748	0.379	0.450	0.628	0.552
LRRNet [9]	0.781	0.923	0.947	0.825	0.760	0.908	0.857	0.449	0.663	0.760	0.415	0.445	0.649	<u>0.564</u>
SHIP [31]	0.792	0.919	0.949	0.823	<u>0.795</u>	0.886	<u>0.861</u>	0.458	0.675	0.751	0.423	0.454	0.621	<u>0.564</u>
DATFuse [12]	0.793	0.914	0.941	0.790	0.733	0.908	0.847	0.461	0.652	0.750	0.390	0.434	0.652	0.556
CrossFuse [22]	0.779	0.915	0.945	0.822	0.781	0.900	0.857	0.446	0.665	0.778	0.380	<u>0.459</u>	<u>0.653</u>	0.563
FusionGAN [13]	0.787	0.938	0.970	0.813	0.709	0.917	0.858	0.453	<u>0.678</u>	0.765	0.397	0.413	<u>0.653</u>	0.560
DDFM [15]	0.794	0.919	0.943	0.813	0.760	0.902	0.855	0.460	0.659	0.749	0.407	0.440	0.644	0.560
Ours	<u>0.800</u>	0.929	0.959	0.831	0.829	0.928	0.879	<u>0.462</u>	0.685	<u>0.769</u>	<u>0.416</u>	0.462	0.655	0.575

multi-modality benchmark (FMB), derived from the M³FD benchmark, which encompasses a wide array of driving scenarios under various lighting conditions. This dataset includes 1,120 image pairs across 14 classes for training and an additional 280 pairs for testing. The experimental setup for semantic segmentation mirrors that of SegMiF [24], with accuracy (ACC) and intersection over union (IoU) serving as the primary metrics.

Fig. 8 presents a qualitative comparison of FreqGAN against seven other competitors. A noticeable observation is that in contexts involving pedestrians and buildings, a single modality image often falls short of yielding precise classifications. Contrastingly, fusion methods generally show marked improvements in semantic segmentation performance. Notably, FreqGAN stands out for its exceptional accuracy in object and scene classifications, closely mirroring the ground truth. The quantitative comparison results are detailed in Table II. The numerical results underscore the significant advancements facilitated by FreqGAN, with it surpassing other SOTA competitors by 9.35% and 13.41% enhancement in mACC and mIoU, respectively. These findings collectively affirm that FreqGAN efficaciously harnesses and amplifies the complementary information inherent in the frequency domain and improves the segmentation performance.

E. Computational Complexity

Table III offers an insightful comparison of computational complexity across various fusion methods, employing metrics such as training parameters (Params.), floating-point operations per second (FLOPs), and average runtime (Time). For this analysis, test images with the size of 256×256 are utilized to ascertain the FLOPs. In this comparative landscape, our FreqGAN exhibits a higher count of training parameters, outperforming most competitors except LRRNet, CrossFuse, and DDFM. This increment in parameters primarily stems from the integration of discrete wavelet transformation within both the generator and dual discriminators, coupled with the implementation of frequency loss functions, naturally inflating the parameter count. However, FreqGAN's efficient multi-level framework and its lightweight architectural design contribute to its noteworthy performance in other complexity aspects. Specifically, it achieves the second rank in terms of FLOPs and runtime, trailing only behind DATFuse. This outcome signifies the balanced computational complexity of FreqGAN, which, despite the inclusion of multiple wavelet transformation modules, maintains a competitive operational efficiency.

F. Ablation Studies

This section introduces ablation studies to assess the effectiveness of integrating frequency components into the

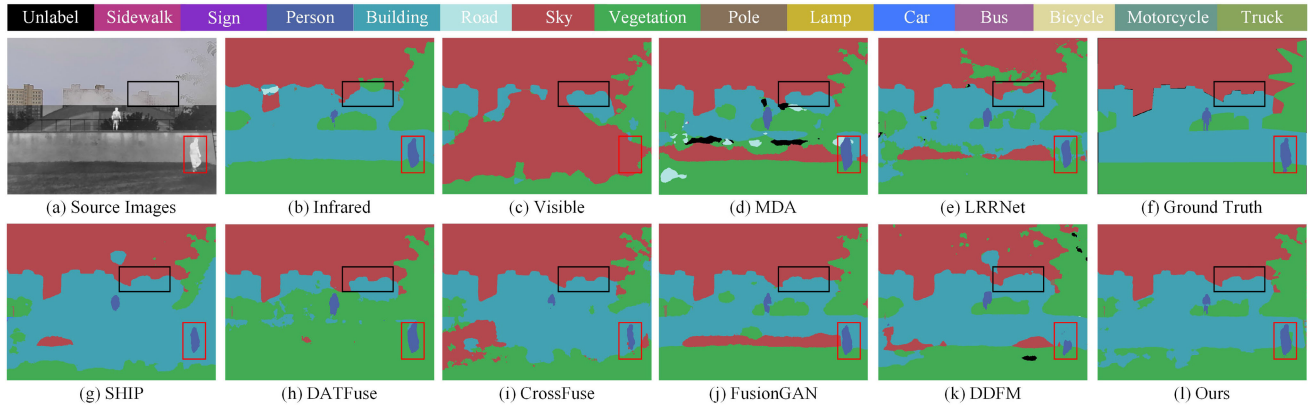


Fig. 8. Qualitative comparisons of segmentation performance for our FreqGAN with the other seven SOTA methods and source images on the FMB benchmark. Note that our FreqGAN effectively classifies objects and scenes with high precision, and its result is close to ground truth.

TABLE II
QUANTITATIVE COMPARISONS OF SEGMENTATION PERFORMANCE ON THE FMB BENCHMARK

Methods	Road		Sidewalk		Lamp		Sign		Vegetation		Sky		Person		Pole		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
Infrared	83.8	79.9	51.4	30.4	70.4	12.2	79.2	54.6	84.6	74.7	95.4	90.2	84.9	63.0	46.1	24.4	74.5	53.7
Visible	84.6	82.7	66.4	32.1	57.4	33.0	83.5	65.0	93.0	81.4	93.5	91.4	84.8	41.1	63.2	37.6	78.3	58.0
MDA [10]	86.4	81.4	48.7	36.5	61.3	41.4	82.8	65.7	88.7	83.2	95.0	93.3	70.1	56.9	73.8	26.6	75.9	60.6
LRRNet [9]	86.1	83.2	47.3	35.8	56.1	35.6	77.3	65.8	89.4	82.2	97.0	94.0	83.2	57.5	57.1	38.5	74.2	61.6
SHIP [31]	86.2	82.0	61.0	33.7	50.6	34.8	75.7	66.5	94.8	80.6	97.6	93.4	80.3	60.8	60.6	36.3	75.8	61.0
DATFuse [12]	85.1	80.0	50.3	21.7	51.4	30.0	84.0	61.5	81.7	78.4	95.6	92.6	77.9	63.1	71.8	39.4	74.7	58.3
CrossFuse [22]	85.5	81.7	81.7	28.0	75.5	35.2	83.1	68.5	91.4	82.7	95.2	92.2	81.4	54.0	63.0	33.7	82.1	59.5
FusionGAN [13]	84.8	80.0	57.8	32.6	50.4	28.5	82.6	61.5	90.4	82.3	93.7	91.3	89.2	62.6	62.1	35.7	76.4	59.3
DDFM [15]	81.2	79.9	53.7	24.0	46.1	31.0	75.4	65.3	87.7	81.2	95.1	91.8	79.0	54.6	49.1	35.1	70.9	57.9
Ours	86.6	84.7	70.2	38.0	73.4	44.1	84.3	70.4	91.2	86.1	97.1	94.6	86.4	69.4	74.9	46.8	83.0	66.8

TABLE III
COMPUTATIONAL COMPLEXITY COMPARISONS OF OUR FREQGAN WITH THE OTHER SEVEN SOTA COMPETITORS

Methods	Params.(M)	FLOPs(G)	Time(s)	
			TNO	M ³ FD
MDA [10]	0.473	36.21	0.153	0.559
LRRNet [9]	7.635	18.23	0.217	0.803
SHIP [31]	0.522	25.24	0.366	0.719
DATFuse [12]	0.011	1.185	0.019	0.047
CrossFuse [22]	22.23	82.02	0.583	0.659
FusionGAN [13]	1.314	57.09	0.513	0.988
DDFM [15]	988.3	2946	59.18	162.1
Ours	1.832	5.535	0.141	0.513

generator, discriminator, and loss function. The TNO benchmark and eight evaluation metrics are employed for qualitative and quantitative comparisons.

1) *Analysis of Frequency-Compensated Generator*: In the generator, we integrate the low and high-frequency components into the encoder and decoder to compensate for extra outlines and details. Meanwhile, we also design a hybrid frequency aggregation module to measure and optimize the activity levels. To verify their effectiveness, we compare the proposed FreqGAN with other models, *i.e.*, lacking DWT

blocks (w/o DWT), removing low-frequency signals (w/o LF), missing high-frequency components (w/o HF), replacing the hybrid frequency aggregation module with two convolutional layers (w/o HFAM), and assigning attentional weights for visible features (β_{vis}^{l-1}). As illustrated in Fig.9, after removing any frequency signal, the fusion outputs fail to control a suitable apparent intensity. Specifically, the fusion models are unable to provide additional target brightness and scene details. Meanwhile, the visual performance of the fusion model without HFAM is similar to that of the model with assigning attentional weights for visible features. This phenomenon demonstrates that a lack of activity level optimization and inappropriate weight assignment degrades the fusion performance. In addition, the quantitative comparisons are presented in Table IV, our FreqGAN achieves the first position across all metrics except for MS-SSIM and SCD. Both qualitative and quantitative experiments illustrate that the design of the frequency-compensated generator is effective.

2) *Analysis of Frequency-Constrained Discriminator*: In dual discriminators, we first adopt DWT to transform input images into frequency components, and then propose a convolutional attention block to refine these frequency signals. Ablation validations are performed on five comparable models, including those without DWT and CABlocks (w/o FA), lacking DWT of dual discriminators (w/o D^W), removing DWT

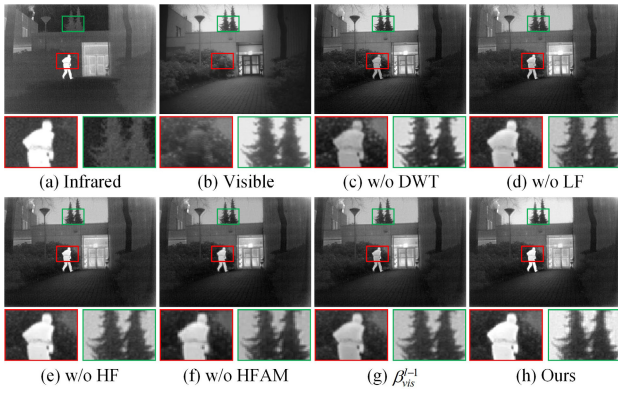


Fig. 9. Visual validations of frequency-compensated generator.

TABLE IV

QUANTITATIVE RESULTS OF FREQUENCY-COMPENSATED GENERATOR

Models	w/o DWT	w/o LF	w/o HF	w/o HFAM	β_{vis}^{l-1}	Ours
AG	<u>4.8456</u>	4.7396	4.4863	3.9895	4.2811	5.0751
SF	9.4482	9.2696	8.5821	<u>9.8759</u>	8.3318	10.203
EN	7.0211	7.0112	<u>7.0266</u>	7.0110	7.0253	7.0287
SD	<u>45.546</u>	42.283	41.336	41.686	41.765	46.060
MS-SSIM	0.9053	0.9089	0.9146	0.9136	0.9338	<u>0.9164</u>
Qw	0.4767	0.5219	<u>0.5229</u>	0.4422	0.5215	0.5241
SCD	1.7186	<u>1.8551</u>	1.7377	1.7993	1.8658	1.7427
VIFF	0.5803	0.5939	0.6072	<u>0.6210</u>	0.5819	0.6350

from single infrared or visible discriminator (w/o D_i^W and w/o D_v^W), and eliminating four CABlocks (w/o CAB). As shown in Fig. 10, after removing one or both DWT and CABlocks, all the models lead to varying degrees of degradation in fusion performance. Specifically, the model without CABlocks fails to control the apparent intensity, while models without D_i^W or D_v^W produce unbalanced results. Furthermore, as depicted in Table V, our FreqGAN achieves the optimal values for metrics AG, SF, EN, SD, MS-SSIM, and Qw, and the suboptimal value for VIFF. These ablation validations demonstrate the rationality of the frequency-constrained discriminator.

3) *Analysis of Frequency-Supervised Loss Function:* We also introduce DWT into the design of the loss function and set parameter r to control the contribution of different frequency components. To assess the influence of different hyperparameters, DWT blocks, and control weights, we compare the proposed FreqGAN with other models, *i.e.*, selecting other parameter settings like SwinFusion [21] ($\lambda_1:\lambda_2:\lambda_3 = 10:20:30$, termed WP1) and AITFuse [23] ($\lambda_1:\lambda_2:\lambda_3 = 1:20:4$, termed WP2), without DWT within the loss function (w/o WLoss), and different control weights $r = \{0.3, 0.5\}$. As shown in Fig. 11, compared with our FreqGAN, the WP1 and WP2 models produce limited target brightness and fuzzy edge details. After removing the DWT block, the fused output cannot maintain the optimal intensity information and texture details. For the control weights r , when it changes from 0.3 to 0.4, the fusion output produces the optimal best visual effect, and when it continues to increase, the fusion outputs show a certain degree of degradation. As shown in Table VI,

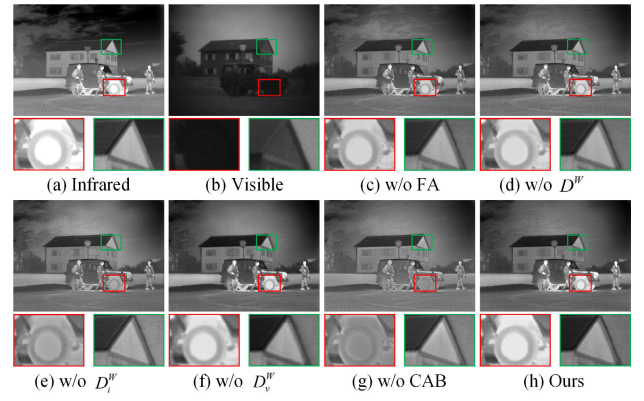


Fig. 10. Visual validations of frequency-constrained discriminator.

TABLE V

QUANTITATIVE RESULTS OF FREQUENCY-CONSTRAINED DISCRIMINATOR

Models	w/o FA	w/o D^W	w/o D_i^W	w/o D_v^W	w/o CAB	Ours
AG	4.2088	4.3723	<u>4.7520</u>	4.6333	4.7056	5.0751
SF	8.0488	8.6566	<u>9.5930</u>	9.3499	9.1240	10.203
EN	7.0188	7.0189	6.9259	<u>7.0197</u>	<u>7.0197</u>	7.0287
SD	41.625	44.416	44.936	<u>45.592</u>	42.248	46.060
MS-SSIM	0.9077	<u>0.9156</u>	0.9149	0.9154	0.9121	0.9164
Qw	<u>0.5226</u>	0.5182	0.5182	0.5169	0.5219	0.5241
SCD	1.8943	1.7418	1.7323	1.7377	<u>1.8017</u>	1.7427
VIFF	0.5830	0.6672	0.6276	0.6284	0.6304	<u>0.6350</u>

we can find that our FreqGAN secures the top ranks for metrics except for EN and Qw, which respectively rank in the third and second position. The ablation experiments prove the effectiveness of the frequency-supervised loss function.

G. Discussion

In the generator, we utilize skip connections to respectively introduce low-frequency signals and high-frequency components into the encoder and decoder, and develop a hybrid frequency aggregation module to progressively optimize the activity levels of various frequency bands. Nevertheless, in the discriminator, we apply the DWT blocks to decompose multiple frequency bands, and these frequency components are directly concatenated. In other words, we adhere to a unified adversarial learning scheme that is unable to independently balance the emphasis on different frequency bands. Hence, we believe that the potential of a divide-and-conquer strategy should be explored, specifically designed to process diverse frequency components both separately and parallelly within the wavelet domain. This prospective innovation allows for a faster and more stable training process, and further enhance the fusion performance.

H. Limitation

Regarding the fusion framework, we incorporate discrete wavelet transformation (DWT) within both the generator and discriminator to segregate encoded spatial features into multiple frequency bands. Although FreqGAN attains generally

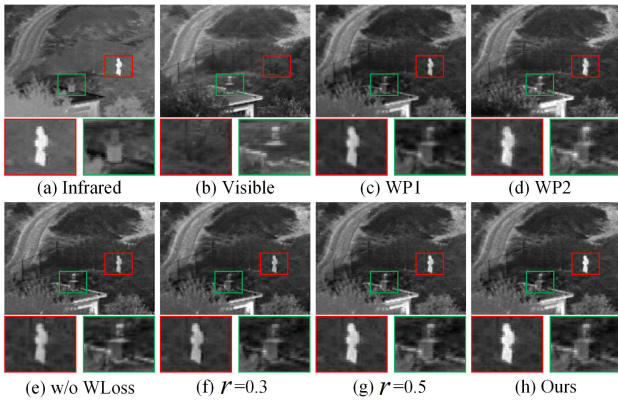


Fig. 11. Visual validations of frequency-supervised loss function.

TABLE VI

QUANTITATIVE RESULTS OF FREQUENCY-SUPERVISED LOSS FUNCTION

Models	WP1	WP2	w/o WLoss	$r = 0.3$	$r = 0.5$	Ours
AG	4.7806	4.9293	4.4548	4.7198	<u>5.0298</u>	5.0751
SF	9.3342	9.7706	8.9230	9.3442	<u>9.8222</u>	10.203
EN	7.1056	6.8778	7.0247	7.0081	<u>7.0501</u>	7.0287
SD	44.053	36.630	45.471	44.819	<u>45.471</u>	46.060
MS-SSIM	0.8989	0.8549	0.9015	0.9117	<u>0.9133</u>	0.9164
Qw	0.5306	0.4932	0.4863	0.5222	0.5228	<u>0.5241</u>
SCD	1.7352	1.4929	1.7089	<u>1.7364</u>	1.7322	1.7427
VIFF	0.5601	0.4690	0.5411	<u>0.6340</u>	0.6222	0.6350

satisfactory performance, it is by no means the most optimal strategy. Fundamentally, the DWT blocks extract frequency information via a fixed, unlearnable transformation operation. In the future, we will explore properly initialized convolutional operations to replace the Haar wavelet kernels. This learnable frequency strategy can hold significant potential for further enhancing the performance of the proposed method.

V. CONCLUSION

In this paper, we unveil FreqGAN, an innovative unified frequency adversarial learning network for infrared and visible image fusion. FreqGAN features a frequency-compensated generator that adeptly manipulates frequency components to refine outlines and augment details. Furthermore, the network incorporates dual frequency-constrained discriminators, which rigorously enforce the generator's capability to reconstruct pivotal frequency information from different modality images. An additional cornerstone of our approach is the implementation of a frequency-supervised loss function. This function, meticulously focused on introducing the frequency components into structural similarity, detail preservation, and intensity control, plays a crucial role in guiding the network's learning process. FreqGAN's effectiveness is rigorously validated against seven other SOTA methods on the TNO and M³FD benchmarks. These comparative assessments unequivocally demonstrate that FreqGAN not only achieves exemplary fusion performance but also maintains a competitive computational complexity. Beyond its core capabilities, FreqGAN

shows promising applications in downstream tasks, particularly enhancing object detection and semantic segmentation.

REFERENCES

- [1] J. Liu, J. Wang, N. Huang, Q. Zhang, and J. Han, "Revisiting modality-specific feature compensation for visible-infrared person re-identification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 10, pp. 7226–7240, Oct. 2022.
- [2] J. Wang, K. Song, Y. Bao, L. Huang, and Y. Yan, "CGFNet: Cross-guided fusion network for RGB-T salient object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 5, pp. 2949–2961, Jul. 2022.
- [3] W. Zhi-She, Y. Feng-Bao, P. Zhi-Hao, C. Lei, and J. Lie, "Multi-sensor image enhanced fusion algorithm based on NSST and top-hat transformation," *Optik*, vol. 126, no. 23, pp. 4184–4190, Dec. 2015, doi: 10.1016/j.jleo.2015.08.118.
- [4] H. Li, X. Wu, and J. Kittler, "MDLatLRR: A novel decomposition method for infrared and visible image fusion," *IEEE Trans. Image Process.*, vol. 29, pp. 4733–4746, 2020.
- [5] Z. Zhao, S. Xu, C. Zhang, J. Liu, and J. Zhang, "Bayesian fusion for infrared and visible images," *Signal Process.*, vol. 177, Dec. 2020, Art. no. 107734.
- [6] Z. Wang, J. Xu, X. Jiang, and X. Yan, "Infrared and visible image fusion via hybrid decomposition of NSCT and morphological sequential toggle operator," *Optik*, vol. 201, Jan. 2020, Art. no. 163497.
- [7] H. Li and X.-J. Wu, "DenseFuse: A fusion approach to infrared and visible images," *IEEE Trans. Image Process.*, vol. 28, no. 5, pp. 2614–2623, May 2019.
- [8] Z. Wang, Y. Wu, J. Wang, J. Xu, and W. Shao, "Res2Fusion: Infrared and visible image fusion based on dense Res2net and double nonlocal attention models," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–12, 2022.
- [9] H. Li, T. Xu, X.-J. Wu, J. Lu, and J. Kittler, "LRRNet: A novel representation learning guided fusion network for infrared and visible images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 11040–11052, Sep. 2023.
- [10] G. Yang, J. Li, H. Lei, and X. Gao, "A multi-scale information integration framework for infrared and visible image fusion," *Neurocomputing*, vol. 600, Oct. 2024, Art. no. 128116.
- [11] Z. Wang, Y. Chen, W. Shao, H. Li, and L. Zhang, "SwinFuse: A residual Swin transformer fusion network for infrared and visible images," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–12, 2022.
- [12] W. Tang, F. He, Y. Liu, Y. Duan, and T. Si, "DATFuse: Infrared and visible image fusion via dual attention transformer," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 7, pp. 3159–3172, Jul. 2023.
- [13] J. Ma, W. Yu, P. Liang, C. Li, and J. Jiang, "FusionGAN: A generative adversarial network for infrared and visible image fusion," *Inf. Fusion*, vol. 48, pp. 11–26, Aug. 2019.
- [14] J. Ma, H. Xu, J. Jiang, X. Mei, and X.-P. Zhang, "DDcGAN: A dual-discriminator conditional generative adversarial network for multi-resolution image fusion," *IEEE Trans. Image Process.*, vol. 29, pp. 4980–4995, 2020.
- [15] Z. Zhao et al., "DDFM: Denoising diffusion model for multi-modality image fusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 8082–8093.
- [16] Z.-Q. J. Xu, Y. Zhang, and Y. Xiao, "Training behavior of deep neural network in frequency domain," in *Proc. Int. Conf. Neural Inf. Process.*, Oct. 2019, p. 26474.
- [17] Z. Wang, J. Wang, Y. Wu, J. Xu, and X. Zhang, "UNFusion: A unified multi-scale densely connected network for infrared and visible image fusion," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 6, pp. 3360–3374, Jun. 2022.
- [18] L. Tang, J. Yuan, H. Zhang, X. Jiang, and J. Ma, "PIAFusion: A progressive infrared and visible image fusion network based on illumination aware," *Inf. Fusion*, vols. 83–84, pp. 79–92, Jul. 2022.
- [19] D. Wang, J. Liu, X. Fan, and R. Liu, "Unsupervised misaligned infrared and visible image fusion via cross-modality image generation and registration," in *Proc. Int. Joint Conf. Artif. Intell.*, Jul. 2022, pp. 3508–3515.
- [20] Z. Zhao et al., "CDDFuse: Correlation-driven dual-branch feature decomposition for multi-modality image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5906–5916.
- [21] J. Ma, L. Tang, F. Fan, J. Huang, X. Mei, and Y. Ma, "SwinFusion: Cross-domain long-range learning for general image fusion via Swin transformer," *IEEE/CAA J. Autom. Sinica*, vol. 9, no. 7, pp. 1200–1217, Jul. 2022.

- [22] H. Li and X.-J. Wu, "CrossFuse: A novel cross attention mechanism based infrared and visible image fusion approach," *Inf. Fusion*, vol. 103, Mar. 2024, Art. no. 102147.
- [23] Z. Wang, F. Yang, J. Sun, J. Xu, F. Yang, and X. Yan, "AITFuse: Infrared and visible image fusion via adaptive interactive transformer learning," *Knowl.-Based Syst.*, vol. 299, Sep. 2024, Art. no. 111949.
- [24] J. Liu et al., "Multi-interactive feature learning and a full-time multi-modality benchmark for image fusion and segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jul. 2023, pp. 8115–8124.
- [25] Y. Yang, J. Liu, S. Huang, W. Wan, W. Wen, and J. Guan, "Infrared and visible image fusion via texture conditional generative adversarial network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 12, pp. 4771–4783, Dec. 2021.
- [26] Z. Wang, W. Shao, Y. Chen, J. Xu, and X. Zhang, "Infrared and visible image fusion via interactive compensatory attention adversarial learning," *IEEE Trans. Multimedia*, vol. 25, pp. 1–13, 2022.
- [27] Z. Wang, W. Shao, Y. Chen, J. Xu, and L. Zhang, "A cross-scale iterative attentional adversarial fusion network for infrared and visible images," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 8, pp. 3677–3688, Aug. 2023.
- [28] J. Liu et al., "Target-aware dual adversarial learning and a multi-scenario multi-modality benchmark to fuse infrared and visible for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5792–5801.
- [29] J. Yao, Y. Zhao, Y. Bu, S. G. Kong, and J. C. W. Chan, "Laplacian pyramid fusion network with hierarchical guidance for infrared and visible image fusion," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 9, pp. 4630–4644, Sep. 2023.
- [30] H. Li, Y. Xiao, C. Cheng, Z. Shen, and X. Song, "DePF: A novel fusion approach based on decomposition pooling for infrared and visible images," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–14, 2023.
- [31] N. Zheng et al., "Probing synergistic high-order interaction in infrared and visible image fusion," in *Proc. IEEE Conf. Comput. Vis. Pattern Recogn. (CVPR)*, Jun. 2024, pp. 26384–26395.
- [32] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
- [33] A. M. Eskicioglu and P. S. Fisher, "Image quality measures and their performance," *IEEE Trans. Commun.*, vol. 43, no. 12, pp. 2959–2965, Dec. 1995.
- [34] W. J. Roberts, J. A. A. Van, and F. Ahmed, "Assessment of image fusion procedures using entropy, image quality, and multispectral classification," *J. Appl. Remote Sens.*, vol. 2, no. 1, pp. 1–28, May 2008.
- [35] Y.-J. Rao, "In-fibre Bragg grating sensors," *Meas. Sci. Technol.*, vol. 8, no. 4, pp. 355–375, 1997.
- [36] K. Ma, K. Zeng, and Z. Wang, "Perceptual quality assessment for multi-exposure image fusion," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3345–3356, Nov. 2015.
- [37] G. Piella and H. Heijmans, "A new quality metric for image fusion," in *Proc. Int. Conf. Image Process.*, Nov. 2003, pp. 173–176.
- [38] V. Aslantas and E. Bendes, "A new image quality metric for image fusion: The sum of the correlations of differences," *Int. J. Electron. Commun.*, vol. 69, no. 12, pp. 1890–1896, Dec. 2015.
- [39] Y. Han, Y. Cai, Y. Cao, and X. Xu, "A new image fusion performance metric based on visual information fidelity," *Inf. Fusion*, vol. 14, no. 2, pp. 127–135, Apr. 2013.
- [40] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.



Zhuoqun Zhang received the B.S. degree in electrical engineering and automation from Tongji Zhejiang College, Jiaxing, China, in 2022. He is currently pursuing the M.S. degree in optoelectronic information engineering with Taiyuan University of Science and Technology, Taiyuan, China. His current research interests include image fusion and deep learning.



Wuqiang Qi received the B.S. degree in software engineering from Taiyuan University of Technology, Taiyuan, China, in 2021. He is currently pursuing the M.S. degree in optical engineering with Taiyuan University of Science and Technology, Taiyuan. His current research interests include image fusion and deep learning.



Fengbao Yang received the Ph.D. degree in measurement technology and instrument from the North University of China, Taiyuan, China, in 2003. From 2004 to 2007, he was a Post-Doctoral Research Fellow with Beijing Institute of Technology. He is currently a Full Professor with the North University of China. His current research interests include information fusion, performance detection and evaluation of complex systems, and information fusion theory of uncertainty.



Zhishe Wang (Member, IEEE) received the B.S. degree in automation from North China Institute of Science and Technology, Taiyuan, China, in 2002, and the M.S. and Ph.D. degrees in signal and information processing from the North University of China, Taiyuan, in 2007 and 2015, respectively. He is currently a Professor with Taiyuan University of Science and Technology. His current research interests include computer vision, pattern recognition, and machine learning.



Jiawei Xu received the M.S. degree in image processing from Hallym University, Chuncheon, South Korea, in 2011, and the Ph.D. degree from the University of Lincoln, Lincoln, U.K., in 2015, with a focus on human factors in driving. He is currently with the Key Laboratory of Intelligent Informatics for Safety and Emergency of Zhejiang Province, Wenzhou University, Wenzhou, China. His research interests include human factors in intelligent driving and aviation.